

GADDE BHANU PRAKASH

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TECHNICAL SKILLS

GenAI & Agentic Systems: LangGraph, LangChain, RAG (Hybrid Search, RRF, BGE Reranker), FastEmbed (BM25), Google Gemini, Ollama (Mistral), Groq (Llama 3.1), Tavily, LangSmith, Qdrant, Prompt Engineering

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, CNN (EfficientNet B4), XGBoost, LightGBM, Prophet, Google OR-Tools (LP), PyOD, Autoencoders, Isolation Forest

Data Engineering & Backend: Python, SQL, FastAPI, PostgreSQL (Neon), SQLAlchemy, Pandas, NumPy, Great Expectations, Docker, CI/CD (GitHub Actions)

Visualization & Tools: Streamlit, Tableau, Power BI, Seaborn, Matplotlib, MLflow, SHAP, A/B Testing, Azure, Render

KEY TECHNICAL PROJECTS

OmniRAG — Multi-Agent Hybrid Research Assistant

[Live Demo](#) —  GitHub

LangGraph · Gemini · Qdrant · FastEmbed · Streamlit · Docker

- Architected a self-correcting multi-agent RAG system using LangGraph; implemented a Critique Agent that iteratively refines outputs, **reducing hallucinations by 90%** and enabling reliable automated research.
- Engineered a hybrid retrieval pipeline combining dense (Gemini 3072-dim) and sparse (BM25) embeddings with Reciprocal Rank Fusion and BGE Cross-Encoder reranking, **boosting retrieval precision by 40%**.
- Resolved Python 3.13 breaking changes by migrating deprecated Google API packages and re-engineering embedding dimensions, ensuring long-term system maintainability and zero-cost deployment via Streamlit Cloud + CI/CD.

SynergyOS — Cognitive Supply Chain Twin & Optimizer

[Live Demo](#) —  GitHub

Prophet · LightGBM · OR-Tools · LangChain · Ollama · FastAPI

- Developed a hybrid forecasting engine (Prophet + LightGBM) integrated with real-time weather APIs for hyper-local demand sensing across 5+ warehouses, enabling accurate 4-week inventory planning.
- Built a multi-objective optimization solver using Google OR-Tools (Linear Programming) to simultaneously minimize holding, stockout, and transfer costs, directly targeting supply chain operational efficiency.
- Integrated a GenAI Co-Pilot (LangChain + Mistral 7B) to translate complex LP optimization results into actionable plain-language alerts, **reducing stakeholder decision-making time by 80%**.

VIGIL — Autonomous AI Data Quality & Observability Platform

[Live Demo](#) —  GitHub

PyTorch · FastAPI · Great Expectations · Gemini · Docker · Render

- Designed an autonomous data healing system utilizing Isolation Forest + PyTorch Autoencoders for anomaly detection, paired with LangChain + Gemini to auto-generate corrective SQL queries.
- Implemented statistical drift detection (KS-Test, Chi-Squared) and 5 automated validation rules via Great Expectations, providing early pipeline failure alerts and a unified Data Trust Score (0-100%).
- Engineered a safety-critic agent to intercept and block destructive SQL commands (DELETE, DROP), ensuring production-safe autonomous data remediation without human intervention.

EXPERIENCE

The Website Makers

Dec 2025 – Mar 2026

Data Analyst Intern

- Automated data validation pipelines for web analytics (GA4, server logs), achieving **100% reporting accuracy** and reducing manual QA overhead by 25%.
- Streamlined cross-functional data collection workflows between engineering and marketing teams, cutting weekly report generation time by 20%.

IBM (1M1B Initiative)

Dec 2025 – Jan 2026

AI Engineering Fellow — Sustainability Track

- Selected for intensive 6-week applied AI fellowship; engineered production-grade RAG and Agentic AI solutions addressing climate data challenges under IBM Labs mentorship.
- Delivered final technical presentation to industry panel, demonstrating scalable LLM architecture design and clear translation of complex AI concepts to non-technical stakeholders.

EDUCATION

IIT Jammu

2026 – Present

Advanced Certification in Data Science with AI

Pursuing

Visvodaya Institute of Technology and Science

2022 – 2026

B.Tech in Computer Science (Artificial Intelligence)

CGPA: 8.0/10.0

CERTIFICATIONS & LEADERSHIP

- **IBM AI + Sustainability Certificate** — Validated expertise in Agentic AI and RAG architectures through official IBM SkillsBuild program.
- **Top 8 in Meghaton Competition (IIIT Hyderabad)**
- **Published Paper:** “Deep Learning – Driven Visual Search and Image Recommendation” (Apr 2026). *IJMTST*, Vol. 12, Issue 04. **G. Bhanu Prakash** et al. UGC Approved (ID: 43137). DOI: 10.5281/zenodo.19324579